

Hibernate Configuration using hibernate.properties File

Steps to Configure Hibernate with MySQL:

1. Create hibernate.properties file:

- The hibernate.properties file contains configuration settings to establish the connection with the MySQL database. The file should be placed in the src/main/resources folder of your project.

The properties in this file:

```
hibernate.connection.driver_class = com.mysql.cj.jdbc.Driver
hibernate.connection.url = jdbc:mysql://localhost:3306/telusko_db
hibernate.connection.username = root
hibernate.connection.password = root123
hibernate.dialect = org.hibernate.dialect.MySQLDialect
hibernate.hbm2ddl.auto = update
hibernate.show_sql = true
hibernate.format_sql = true
```

2. Code to Automatically Identify hibernate.properties:

- In the Java code, you don't need to manually configure each setting using config.setProperty(). Hibernate can automatically detect the hibernate.properties file and use it to configure the database connection.

```
public static void main(String[] args) {
    Configuration config = new Configuration();
    SessionFactory sessionFactory = null;
    Session session = null;
    Transaction transaction = null;
```

```

    try {
        // Load hibernate.properties automatically
        config.configure();
        config.addAnnotatedClass(Student.class);

        sessionFactory = config.buildSessionFactory();
        session = sessionFactory.openSession();

        transaction = session.beginTransaction();

        // Perform your database operations here

        transaction.commit();
    } catch (Exception e) {
        if (transaction != null) {
            transaction.rollback();
        }
        e.printStackTrace();
    } finally {
        if (session != null) {
            session.close();
        }
    }
}

```

- `config.configure();` loads the Hibernate settings from the `hibernate.properties` file.
- `config.addAnnotatedClass(Student.class);` tells Hibernate to look for entity annotations in the `Student` class.

2.Key Concepts:

- **hibernate.connection.driver_class:** Specifies the JDBC driver class.
- **hibernate.connection.url:** The database URL to connect to.

- **hibernate.connection.username/password:** Credentials for the database connection.
- **hibernate.dialect:** Hibernate uses this to generate SQL optimized for the specific database.
- **hibernate.hbm2ddl.auto:** Manages the schema update (update, create, create-drop, validate).
- **hibernate.show_sql:** If set to true, Hibernate will log the executed SQL queries.
- **hibernate.format_sql:** Pretty-print the SQL queries.